

APPROXIMATE COSMIC SIZE TO EARLY ESTIMATE WEB APPLICATION DEVELOPMENT EFFORT

Lucia De Marco Filomena Ferrucci **Carmine Gravino**,
University of Salerno

Outline

- Motivation
 - Using COSMIC-based approximate countings
- Empirical study
 - Design
 - Results and Discussion
- Future work

Motivation



- FSM methods widely applied in software engineering for estimating the size of software systems
- Few investigations in the context of Web applications:
 - highlighted difficulties in the application of FPA (1st generation of FSM method) suggesting the use of other FSM methods such as COSMIC (2nd generation)
 - Encouraging results in the use of COSMIC in previous studies (SEAA'08, SEAA'09, SEAA'11)

Motivation (2)



- We analyzed approximate counting to early estimate effort of Web applications
- “early sizing”
 - Early in the life of a project before the FURs have been specified in detail
- “rapid sizing”
 - when a measurement is needed but there is insufficient time or resources to apply the standard detailed method
- These motivations are not mutually exclusive and contribute to reach a trade-off between a correct measurement and time and budget available

Definition



- number of COSMIC Functional Processes (CFunP)
 - determining the number of Function Processes by applying the standard COSMIC method
- Average Functional Process (AFP) suggested by the COSMIC community
 - 1st step: the functional processes (FPs) of previously developed applications are identified. The size for each FP is measured in terms of CFP of COSMIC
 - 2nd step, FPs of a new application are determined. The approximate size is calculated by multiplying the number of FPs and the average size for a FP obtained in 1st step

Research questions



- RQ1: Can CFunP and AFP provide good early size estimations for Web effort prediction?
- RQ2: Are the early effort estimations obtained using CFunP and AFP statistically better than the estimations obtained using baseline benchmarks?
- RQ3: Are the early effort estimations obtained using CFunP and AFP comparable to the estimations obtained using the standard COSMIC method?

Data set



- Data for our study provided by an Italian software company
 - Specialized in the design, development, and management of solutions for Web portals, enterprise intranet/extranet applications (such as Content Management Systems, e-commerce, work-flow managers, etc.), and GIS.
 - certified ISO 9001:2000, and also a certified partner of Microsoft, Oracle, and ESRI
- The company provided us a set of 25 Web applications
 - e.g., e-government, e-banking, Web portals, and Intranet applications
 - developed by exploiting a wide range of Web-oriented technologies, such as J2EE, ASP.NET, etc.

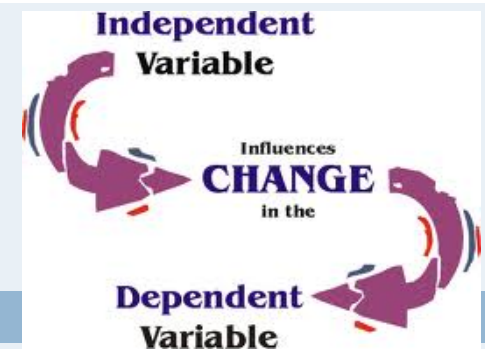
Data set & variables



- Company exploit timesheets to keep track of the Web application development effort
- A questionnaire to collect info to calculate size measures (i.e., COSMIC, CFuncP, AFP)
- COSMIC Measurement Manual (version 2.2.)
 - One of the authors analyzed the filled questionnaires and the analysis and design documents, to cross-check the provided information. The same author calculated the values of the size measures

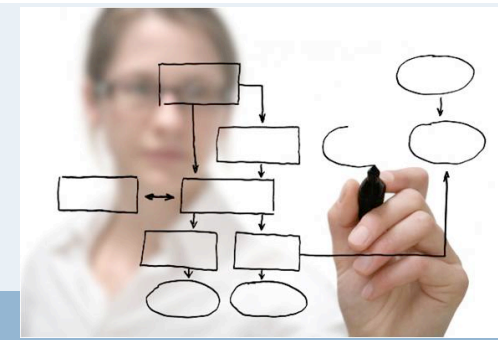
Var	Obs	Min	Max	Mean	Median	Std. Dev.
EFF	25	1176	3712	2577	2686	988.14
CFP	25	163	1090	602	611	268.47
CFPafp	25	183	1417	610.8	545	299.68
nCFunP	25	23	177	77	69	37.56

Estimation technique

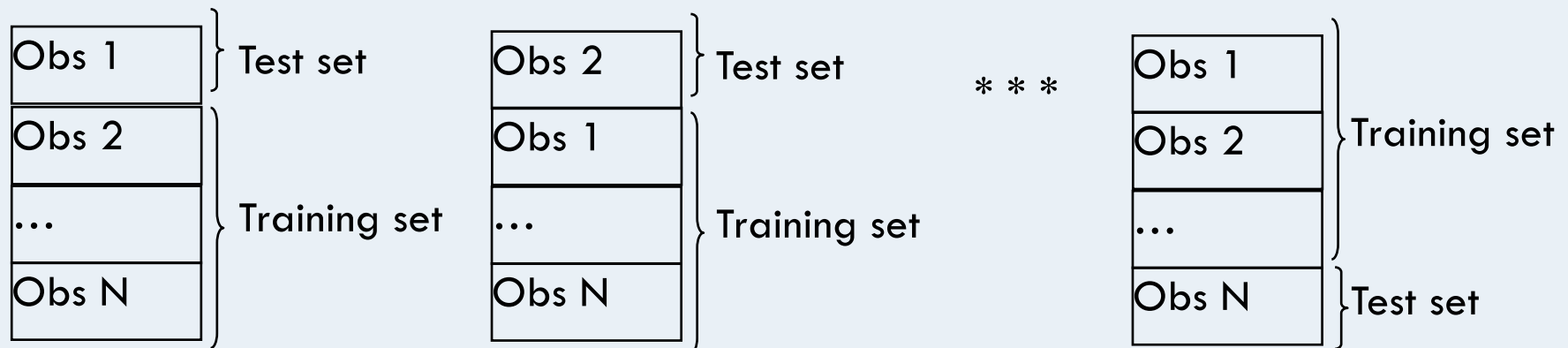


- Simple Linear Regression (SLR)
 - $EFF = a * CFP + b$ / $EFF = a * nCFunP + b$ / $EFF = a * CFP_{afp} + b$
 - R^2 to evaluate the goodness of fit, t-statistics for the statistical significance of the independent variables
- Baseline benchmarks
 - the mean of effort (MeanEFF) and the median of effort (MedEFF) of the past developed applications as predictor of the effort
 - To verify whether for a software company it could be more useful to simply use MeanEFF or MedEFF to have an early estimation of the size rather than dealing with more complex methods

Validation method



- Leave-1-out cross validation
 - $N=25$ different subsets of training (TS) and validation (VS) sets, where each validation set had one observation. TS used to build the estimation model with SLR and VS used to assess the obtained estimates



Evaluation criteria



- MMRE & MdMRE: Mean & Median of Magnitude of Relative Error

$$MRE = | \text{Actual effort} - \text{Predicted effort} | / \text{Actual effort}$$

- An MMRE less than 0.25 (i.e., 25%) acceptable (Conte et al.)
- $\text{Pred}(0.25) = k/N$
 - k is the number of obs with $MRE \leq 0.25$
 - A $\text{Pred}(0.25) > 0.75$ (i.e., 75%) acceptable (Conte et al.)
more than 75% of the estimated values fall within 25% of their actual values
- MAR & MdAR: Mean & Median of Absolute Residuals (AR)

$$AR = | \text{Actual effort} - \text{Predicted effort} |$$

Results: estimation models



nCFunP based model

Variable	Value	Std.Err	t-value	p-value
Coefficient	23.66	204.83	3.69	<0.01
Intercept	755.46	2.4	9.86	<0.01

R ²	Adjusted R ²	Std. Err	F	Sign. F
0.81	0.8	441.6	97.15	<0.01

$$EFF = 23.66 * nCFunP + 755.46$$

$$EFF = 2.96 * CFP_{afp} + 771.8$$

$$EFF = 3.43 * CFP + 512.43$$

CFP_{afp} based model

Variable	Value	Std. Err	t-value	p-value
Coefficient	2.96	0.31	9.69	< 0.01
Intercept	771.8	206.63	3.74	< 0.01

R ²	Adjusted R ²	Std. Err	F	Sign. F
0.803	0.79	448	94	< 0.01

CFP based model

Variable	Value	Std. Err	t-value	p-value
Coefficient	3.43	0.28	12.3	< 0.01
Intercept	512.43	183.14	2.8	0.01

R ²	Adjusted R ²	Std. Err	F	Sign. F
0.87	0.86	366.6	151.3	< 0.01

- High R² and F values and Sign.F <0.05: the prediction is indeed possible with a high degree of confidence
- Significant predictors with a confidence of 95% (p-value<0.05)

Results: RQ1 and RQ2



Approach	MMRE	MdMRE	Pred(25)	MAR	MdAR
nCFunP	0.17	0.15	0.8	379.79	374
CFPaftp	0.19	0.17	0.8	386.6	393
CFP	0.13	0.07	0.88	275.64	198
MeanEFF	0.45	0.31	0.44	842.89	934
MedEFF	0.47	0.31	0.44	904.44	1010

- (RQ1) *CFunP and AFP provide good early size estimations for Web effort estimation*
 - $MMRE < 0.25$ and $Pred(0.75) > 0.75$
- (RQ2) *The early estimations obtained with CFunP and AFP are statistically better than the estimations of baseline benchmarks*
 - MAR and MdAR values of nCFunP and CFPaftp two times less than those of MeanEFF and MedEFF
 - a large effect size, i.e., $r > 0.57$ for nCFunP vs MeanEFF, nCFunP vs MedEFF, and for CFPaftp vs MeanEFF, CFPaftp vs MedEFF

Results: RQ3



Approach	MMRE	MdMRE	Pred(25)	MAR	MdAR
nCFunP	0.17	0.15	0.8	379.79	374
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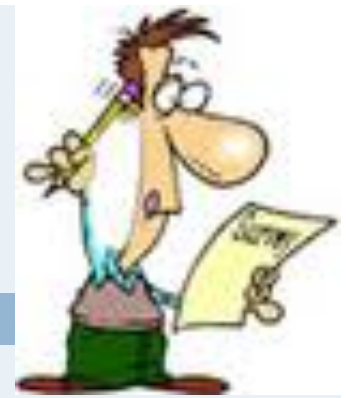
- MMRE, MdMRE, Pred(25), and MAR and MdAR values of CFP based model are better than those achieved with nCFunP and CFPaftp based models
- ARs obtained with CFP based models are statistically significant less than those achieved with nCFunP and CFPaftp based models (p-values = 0.02) with a medium effective size ($r=0.41$ for CFP vs nCFunP and $r=0.42$ for CFP vs CFPaftp)
- We cannot positively answer RQ3

Main findings



- CFunP and AFP can be profitably exploited to obtain early estimations of effort for the Web applications
 - nCFunP based models provided better estimates than AFP based models
- the effort estimations obtained with the size calculated by using the standard COSMIC method are significant better
 - Software company can start to estimate the size in terms of CFunP in the early phase
 - Then, it has to adopt the standard COSMIC to achieve more accurate effort estimations to be employed in the later phases of the development process

Threats to validity



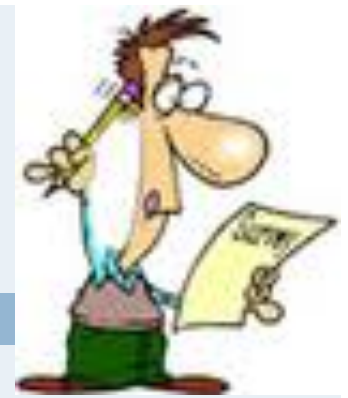
Internal validity

- participants' authoring and designing experience, and reliability of the data and lack of standardization. Our best to control “extraneous factors”: use of the same questionnaires for all the Web applications and the project managers were instructed on how to use the questionnaires

External validity

- Web applications involved in study are representative samples of modern Web applications, taking into account their type, functionality, target platforms, and complexity
- Replications in other industrial contexts

Threats to validity (2)



Construct validity

- The COSMIC based measures are well known and used
- The collection of the information to determine size measures and actual effort represents a crucial aspect

Conclusion validity

- Evaluation criteria allowed assessing in an objective way the effectiveness of the predictions
- We carefully applied the estimation methods and the statistical tests, verifying all the required assumptions